DOC313 & DOC314: Examination Compendium (2022)

Mathematics for Computing & Digital Skills

Part 1: DOC313 Mathematics for Computing

Algebra and Simplification

Question 1 When $(x-1)^6$ is expanded, identify the coefficients for each term:

- ☐ The coefficient of x^6 is: ___________
- ☐ The coefficient of x^5 is: ___________
- ☐ The coefficient of x^4 is: ___________
- ☐ The coefficient of x^3 is: ___________
- ☐ The coefficient of x^2 is: ___________
- ☐ The coefficient of x^1 is: ___________
- ☐ The Constant value is: ___________

Question 9 & 14 Simplify the following expressions as much as possible. Write the final answer with positive powers.

1. $\dfrac{(2ab^2c)^2}{a^3b^2c^3}$
2. $\dfrac{x^6y^6}{2^4y^{-1}x^2}$

Calculus and Sequences

Question 5

(a) Calculate the gradient of $f(x) = x^2 + 6x + 1$ at $x = 3$.

(b) Calculate the Area below the graph $f(x) = 3x^2 - 4x + 5$ from $x = 2$ to $x = 4$.

Question 9 & 12

1. Find the 15^{th} value of the arithmetic sequence: $4, 12, 20, 28, \ldots$
2. Find the sum of the first 10 values of the geometric sequence: $18, 36, 72, 144, \ldots$

Question 7 Find the result of the summation:

$$\sum_{r=1}^{3}(r^2+2)$$

Equations and Logarithms

Question 8, 10, & 11 Solve the following quadratic equations:

1. $x^2 - 5x - 14 = 0$
2. $8x^2 - 22x + 12 = 0$ (Identify factors and solutions).
3. $x^2 + 4x - 6 = 0$ (By the method of **Completing the Square**).

Question 13 Solve for x:

$$\log_2(6x) - \log_2(x-3) = 3$$

Discrete Math and Number Systems

Question 8 Given the matrices:

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 4 & 1 & 1 \\ 5 & 1 & 1 \end{bmatrix}$$

Calculate the product $A \cdot B^T$.

Question 7 How many different ways can the characters in the word **'WELLNESS'** be arranged so that the 'W' and 'N' are always together and also both 'S' characters are also together?

Question 10 Convert the decimal value 59.6875 to:

1. Binary
2. Octal
3. Hexadecimal

Part 2: Algorithm Logic and Digital Skills

Algorithm Development

Question 2 Complete the algorithm below to enter 5 integer numbers and display the total count of positive and negative integers.

1. Start
2. Initialize `positivecount` to ______, `negativecount` to 0, and `counter` to ______.
3. Ask for an integer number.
4. If the entered number is > 0 then increase `positivecount` by 1, else increase `negativecount` by ______.
5. Increase `counter` by ______.
6. If `counter` is less than ______, then go to step 3.
7. Display `positivecount` and `negativecount`.
8. Stop.

Digital Skills (DOC314)

Question 28 The basic architecture of the computer was developed by:

- ◯ Gordon Moore
- ◯ Charles Babbage
- ◯ John Von Neumann
- ◯ Blaise Pascal